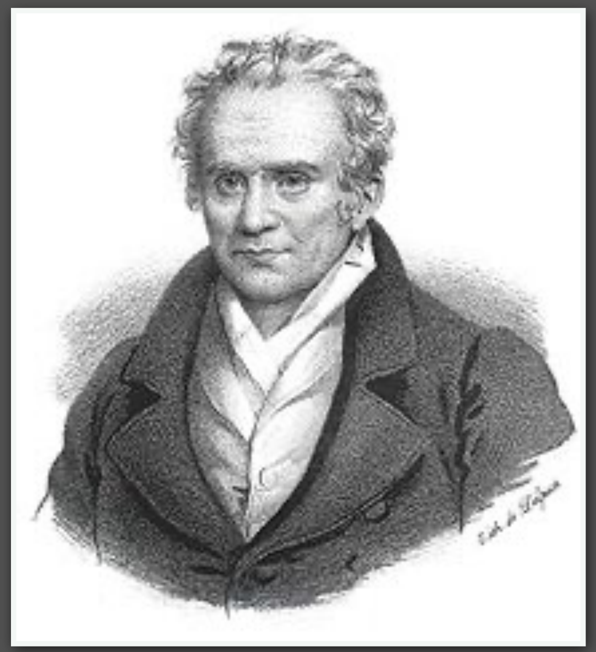
Last Time

- **Measures**: a formal, general way to handle data of various kinds
- **Push-forwards:** generalizing the notion of correspondence/matching
- **Monge Formulation**: discrete (=assignment) and continuous (=mapping)
- **Brenier's Theorem**: Simple conditions that guarantee solution to Monge

Last Time



Monge's Problem:
(Matching/permutation version)

$$\min_{\sigma \in \Sigma_n} \frac{1}{n} \sum_{i=1}^{n} \mathbf{C}_{i,\sigma(i)}$$

Monge's Problem:
(Discrete measure version)

$$\min_{T_\sharp \alpha = \beta} \frac{1}{n} \sum_{i=1}^{n} c(x_i, T(x_i))$$

Monge's Problem:
(Continuous / general version)

$$\min_{\beta = T_\sharp \alpha} \int_X c(x, T(x)) \mathrm{d}\alpha(x)$$

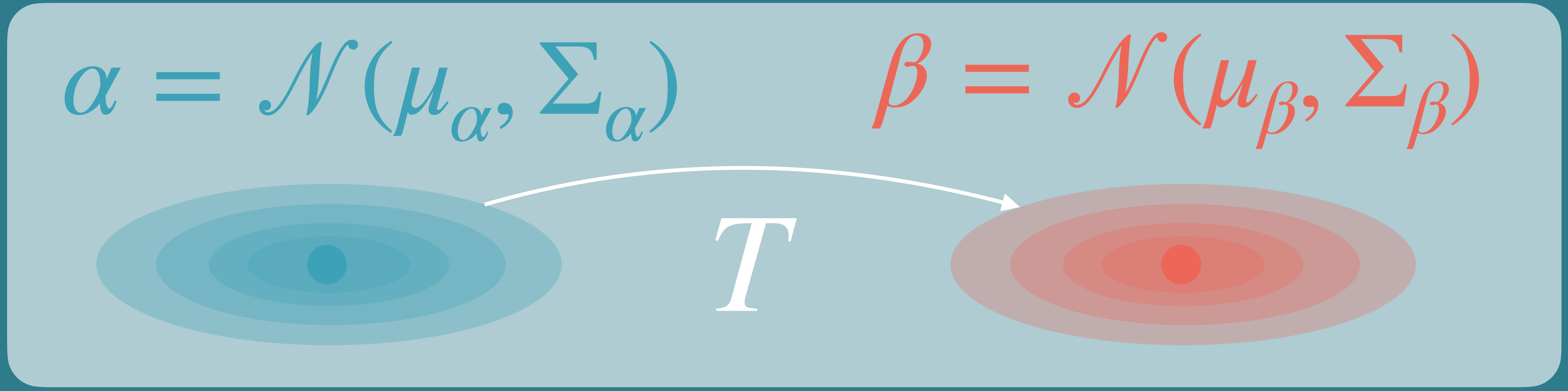
Theorem [Brenier, 1991]:

Assumptions: $c(x, y) = \|x - y\|^2$, α has a density $\rho_\alpha = \frac{\mathrm{d}\alpha}{\mathrm{d}x}$

Then:

- Monge problem has a **unique** solution T
- T is the gradient of a convex function

OT between Gaussians has closed form



$$T(x) = \mu_\beta + A(x - \mu_\alpha); \; A = \Sigma_\alpha^{-\frac{1}{2}} (\Sigma_\alpha^{\frac{1}{2}} \Sigma_\beta \Sigma_\alpha^{\frac{1}{2}})^{\frac{1}{2}} \Sigma_\alpha^{-\frac{1}{2}}$$